
WHITE PAPER · PORTFOLIO & AI GOVERNANCE

AI Can Grow Only as Fast as People Can Supervise It

Every agent adds output. It also adds work that someone must review and own.

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EXECUTIVE SUMMARY

Count the human review work before adding another AI agent.

Every AI agent adds output. It also adds work for people who check important results, handle exceptions, and answer when the AI is wrong. Agents can produce work in seconds, but human attention does not grow at the same speed. This paper shows leaders how to count that work before ownership becomes a name on a chart.

- 01** Ask one named owner to explain what the agent did, what was checked, what failed, and what changed.
- 02** Match review to risk. Sample easy, low-risk work and require a person for high-risk or unusual cases.
- 03** Count review time as a cost in every funding choice. Often the first move is to shift reviewer time from stable, low-risk work to higher-risk work before adding more AI.

The hidden limit

You are told to add more AI agents, find more uses, and get more work done. The plan may count model cost, data, and technical skill. It may not ask a harder question: How many AI decisions can your people still review, explain, and stand behind?

That question sets a real limit. An agent can create output in seconds. A person still needs time and judgment to check important results, handle exceptions, and answer when the AI is wrong. If the number of agents grows faster than that human work, the company does not gain control. It gains work that no one can fully own.

McKinsey expects human oversight to limit how widely companies can use AI agents. It also reports that teams of two to five people can already supervise 50 to 100 specialized agents.^[1] That figure is an early report, not a safe benchmark for every company. Still, it points to the question leaders need to answer: Where is the limit for this team and this work?

A company may not notice the limit when it crosses it. The warning may look like an owner who cannot explain what an agent did, a review that became a quick approval,

or an incident from a system no one was watching closely. By then, the gap has already become an operating risk.

What the team runs out of

The scarce resource is human oversight: the time and judgment people use to review AI output, own decisions, and respond to failures. It has three traits.

First, it is finite. One person can own only so many decisions before ownership becomes a name on a chart. Second, it is shared. A new agent may need help from security, legal, risk, and operations. Those experts also support other systems. Third, it is used whether anyone counts it or not.

That last trait makes the problem easy to miss. A company can add agents without adding the people, tools, or time needed to supervise them. Output rises at once. The oversight gap appears later, when a hard case arrives or a failure reaches another team.

Here is the shift: model quality may open the door, but human oversight decides how much AI the organization can keep live. Leaders can manage that limit once they treat it as work that can be named, measured, and funded.

Put a number on supervision

Start with the span of supervision: how many agents, decisions, or units of AI work one accountable person can genuinely own. There is no single number for every workflow. The span gets smaller when errors are costly, hard to see, or hard to reverse. It can be larger when errors are easy to spot and fix.

Research on human control of automated systems shows why the span has a limit. In one simulation, a person supervised several semi-autonomous aircraft. Some operators reached cognitive saturation and changed how they managed the work. A management-by-exception aid also produced signs of automation bias.[6] In another simulation, 42 U.S. Navy personnel supervised 8, 12, or 16 highly automated missiles. Performance and awareness fell at 16.[7] These studies do not set a limit for business AI. They show that one person's attention can become the weak point even when the machines act with some independence.

Estimate the span for each workflow. Ask four questions:

- How much harm can one wrong result cause?
- How hard is an error to detect?
- How often does the agent meet a case it has not seen?
- How much time does the owner need to review exceptions and follow up?

Then use a weekly test: Can the named owner explain what the agent did, what was checked, what failed, and what changed? If the honest answer is no, the team has already taken on more than it can supervise.

Review by risk, not every action

A person need not read every output. That would place all AI work back at human speed. The goal is to use attention where an error matters most.

Low-risk, easy-to-check work may need sampling. High-risk work may need a person to approve each case. New or uncertain work may need a tighter review until the team understands its typical failures. The rules should say what is sampled, what always gets a human check, what triggers a deeper review, and who can stop the workflow.

McKinsey describes people moving “above the loop”: they set policy, watch unusual cases, and change how involved they are instead of checking every action.[1] This approach can stretch a fixed amount of attention. It works only when the review rules are written and the exceptions reach someone who can act.

Use automation to extend reviewers

Technology can help people supervise more work. McKinsey describes critic agents that challenge output, guardrail agents that enforce policy, and compliance agents that watch rules.[1] Deloitte describes agents watching other agents, with people watching the control layer.[2]

Public standards add practical controls. The NIST Generative AI Profile suggests ongoing monitoring, red-team exercises, independent evaluation, incident response, and structured feedback.[3] The NIST AI RMF Core places this work in a continuing cycle of govern, map, measure, and manage.[4]

These controls can find common errors and send unusual cases to people. They do not remove human ownership. A critic agent can miss a problem. A guardrail can be too broad or too weak. The control layer also needs tests, monitoring, and someone who can change or stop it.

Use automation as a multiplier for a human team. Count the work needed to maintain that automation as part of the review load.

Treat oversight as shared capacity

Every live AI system draws on a shared group of reviewers and experts. That makes oversight a portfolio choice. A new agent may look useful on its own while still taking time away from a more important system.

Each funding request should state its expected review load. It should name the owner and the type of review. It should also list the experts needed, the main warning signs, and the signals that would pause or stop the work. The portfolio owner can then compare that demand with the people who are actually available.

When the pool is full, the first move should often be reallocation. A stable, low-risk workflow may need less review than it did at launch. Those hours can move to a new or high-risk system. Better allocation can support more useful AI without hiring another reviewer.

More model features do not solve a shortage of supervision. Once people are the limit, leaders should compare the next model expense with monitoring, evaluation, workflow design, and reviewer time.

Watch for three warning signs

The first warning is ownership in name only. The system has an owner on paper, but that person cannot say what it did this week or which exceptions remain open.

EXHIBIT 1**Signs that supervision is too thin, and the response****Owner in name only**

The owner cannot explain what the agent did this week.

Rubber-stamp review

A person approves work without enough time or evidence to inspect it.

Surprise incidents

Failures appear in systems no one was watching closely.

The response

Narrow the work, pause new uses, or add review before adding AI.

The second warning is rubber-stamp review. A person approves output without enough time or evidence to inspect it. The approval creates a record of control that did not really happen.

The third warning is surprise. A failure appears in a part of the AI estate that no one was watching closely. The incident is often treated as a model problem even when the deeper cause was thin supervision.

When these signs appear, slow something down. Reduce the number of live cases, narrow the agent's role, pause new uses, or add review before adding more AI. A better model may help later, but it does not replace the missing owner.

Keep the people who can catch the error

Oversight comes from specific people who know the work. They can tell when an answer looks polished but is wrong for the case. That judgment takes time to build.

A company can weaken its own controls when it cuts experienced staff to fund AI. The work may appear to move faster, but fewer people remain who can test an unusual result or understand its effect on customers, employees, or operations. The company adds AI at the same time that it removes the people who can safely own it.

Reviewer time belongs in the workforce plan. Leaders should identify which skills each workflow needs, how many trained owners are available, and how new owners will learn. This is part of the cost of running AI, not spare effort that appears after launch.

What to do next

Choose one live AI workflow and build a simple supervision record:

EXHIBIT 2

A simple supervision record

1. Name the owner

One person answers for the AI agent's results.

2. Set the review rule

Say what is checked, sampled, or sent for deeper review.

3. Count the time

Track review and follow-up work each week.

4. Name the warning signs

Write the events that trigger a deeper review.

5. Name the experts

List who the workflow may need for hard cases.

6. Set a stopping point

Know when to narrow, pause, or stop the AI workflow.

- Name the person who owns the result.
- List the output that receives a human check and the output that is sampled.
- Record the warning signs that trigger a deeper review.
- Estimate the owner's weekly review and follow-up time.
- Name the experts the workflow may need for hard cases.
- Set the point at which the team will narrow, pause, or stop the work.

Review the record after four weeks. Compare the expected load with the time people actually spent. Check whether exceptions reached the right person and whether the owner can explain what changed. Use that evidence to adjust the span before another agent goes live.

Two limits keep the claim honest

The McKinsey figure of two to five people for 50 to 100 agents is not a universal ratio. [1] Risk, task design, and the cost of errors can change the safe span. The human-factors studies also come from controlled settings outside enterprise AI. [5][6][7] They support the need to count attention, but they do not tell a company its number.

The return on spending more for oversight is also not well measured. NIST, McKinsey, and Deloitte describe useful practices, but they do not provide a formula for the right staffing level. [1][2][3][4] Start with a cautious estimate. Measure the real load, missed issues, and response time. Then change the span with evidence.

The choice leaders control

More agents can raise output for a while. The harder test is whether named people can still explain the work, catch important errors, and act when something fails.

Count that human work before approving the next agent. Fund it as part of the system. Move it toward the highest risk. Keep the experienced people who supply it. AI can grow only as fast as the organization can still answer for what it does.

Notes and sources

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About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

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